

MTL704 - 17th April

Frank - Wolf

(P) $\min f(x)$
subject to $x \in S = \{x \in \mathbb{R}^n \mid \begin{matrix} Ax=b \\ x \geq 0 \end{matrix}\}$

$A: m \times n$
 $b: m \times 1$

non empty
& compact

f is convex and differentiable

(P) is a convex program
 $\Rightarrow$ KKT is Necessary and sufficient
for optimality

(P_k) $\left\{ \begin{array}{l} x^0 \in S \text{ arbitrary} \\ \min \quad 0^T x \\ \text{s.t.} \quad Ax=b \\ \quad x \geq 0 \end{array} \right\} \text{ LP}$

$\bar{x}_k$ is its
sol

At any iteration k , we have $x_k^* \in S$

$$\begin{aligned} f(x) &\cong f(x_k) + (x - x_k)^T \nabla f(x_k) \\ &= \underbrace{f(x_k) - x_k^T \nabla f(x_k) + x^T \nabla f(x_k)}_{\text{constant scalar}} \end{aligned}$$

$$w(\bar{x}_k) \leq w(x_k)$$

$$|w(\bar{x}_k) - w(x_k)| < \varepsilon$$

$$w(\bar{x}_k) < w(x_k)$$

$$w(\bar{x}_k) = w(x_k)$$

$$\text{let } d_k = \bar{x}_k - x_k$$

$$\begin{aligned} d_k^T \nabla f(x_k) &= (\bar{x}_k - x_k)^T \nabla f(x_k) \\ &= w(\bar{x}_k) - w(x_k) \\ &< 0 \end{aligned}$$

claim:
Stop
as x_k itself
is optimal
sol for (P)

$$x_{k+1} = x_k + \alpha d_k$$

$$0 < \alpha \leq 1$$

$$\left[\begin{array}{l} x^k \in S \\ \bar{x}_k \in S \\ S \text{ is convex} \\ \therefore x_{k+1} \in S \end{array} \right]$$

show x_k
is KKT pt
for (P)

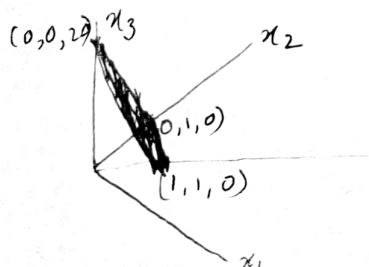
we have $\bar{x}_k$ is a KKT pt for (P)

~~we have $\bar{x}_k$ is a KKT pt for~~

$$L(x, \lambda, \mu) = x^T \nabla f(x_k) + \lambda^T (Ax - b) - \mu^T x$$

$$\lambda \in \mathbb{R}^m, \mu \in \mathbb{R}^n, \mu \geq 0$$

Eg. $\min f(x_1, x_2, x_3) = 100x_1^2 + x_2^2 + (x_3 - 20)^2$
subject to $S \triangleq$ simplex in 3D with vertices
 $(0, 0, 20)$, $(0, 1, 0)$ and $(1, 1, 0)$



$$\nabla f(x) = \begin{pmatrix} 200x_1 \\ 2x_2 \\ 2(x_3 - 20) \end{pmatrix} \quad \swarrow \text{solution}$$

Ans is $(0, 0, 20)^T \in S$

$$\nabla f((0, 0, 20)^T) = 0$$

let $x_0 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ be starting pt. in S

$$\min_{x \in S} w(x) = \nabla f(x_0)^T x$$

$$ST \quad x \in S$$

$$\equiv \min_{ST \quad x \in S} 200x_1 + 2x_2 - 40x_3$$

for LPP sol are @ extreme pt.

$\therefore$ Sol of this LP must be one of the vertex of S .

solⁿ $\bar{x}_0$ is $(0, 0, 20)^T$

$$w(\bar{x}_0) = -800 < w(x_0) = 200$$

$$d_0 = \bar{x}_0 - x_0 = \begin{pmatrix} -1 \\ -1 \\ 20 \end{pmatrix}$$

$$x_1 = x_0 + \alpha_0 d_0 = \begin{pmatrix} 1 - \alpha_0 \\ 1 - \alpha_0 \\ 20\alpha_0 \end{pmatrix}$$

α_0 is a s.d. of

$$\min_{0 < \alpha \leq 1} \phi(\alpha) = f(x^0 + \alpha d_0)$$

$$\min_{0 < \alpha_0 \leq 1} 100(1-\alpha_0)^2 + (1-\alpha_0)^2 + (20\alpha_0 - 20)^2$$

$$= \underbrace{(1-\alpha_0)^2}_{\substack{\uparrow \\ \min = 0 \\ \text{when } \underline{\alpha_0 = 1}}} (100 + 1 + 400) \quad \uparrow \text{const}$$

$$x_1 = \begin{pmatrix} 0 \\ 0 \\ 20 \end{pmatrix} \quad \nabla f(x_1) = 0$$

Theorem: (P) $\min f(x)$
s.t. $x \in S = \{x \in \mathbb{R}^n \mid Ax = b, x \geq 0\}$

- (i) S is non-empty and compact
 $\left\{ \begin{array}{l} \text{closed} \\ \text{bounded} \end{array} \right.$

(ii) $f \in S$, convex, ∇f is Lipschitz

the seq $\{x_k\}$ generated by Frank Wolfe method converges to the KKT pt or solution x^* of (P).

proof

$$\omega(\bar{x}_k) \leq \omega(x_k)$$

$$\Rightarrow x_k^T \nabla f(x_k) \leq x_k^T \nabla f(x_k)$$

$$\Rightarrow (x_k - \bar{x}_k)^T \nabla f(x_k) \geq 0$$

$$\Rightarrow g_k = (x_k - \bar{x}_k)^T \nabla f(x_k)$$

$$\Rightarrow g_k \geq 0$$

$$\text{Claim: } f(x_k) - f(x^*) \leq g_k$$

Note: $x_k \in S$ and x^* is optimal sol of (P)
 $\Rightarrow f(x^*) \leq f(x_k)$

as f is convex on S and $x_k \in S$

$$f(y) - f(x_k) \geq (y - x_k)^T \nabla f(x_k) \quad \forall y \in S$$

In particular set $y = x^* \in S$

$$f(x^*) - f(x_k) \geq (x^* - x_k)^T \nabla f(x_k)$$

$$\Rightarrow f(x_k) - f(x^*) \leq (x_k - x^*)^T \nabla f(x_k)$$

Note that $\bar{x}_k$ is a solⁿ of (P_k) and $x^* \in S$

$$\omega(\bar{x}_k) \leq \omega(x^*)$$

$$-\nabla f(x_k)^T \bar{x}_k \geq -\nabla f(x_k)^T x^*$$

$$f(x_k) - f(x^*)$$

$$\leq f(x_k - \bar{x}_k)^T \nabla f(x_k) = g_k$$

proving the claim.

Next f is L -smooth (∇f is Lipschitz)
 for Lipschitz and constant function we have
 descent lemma

$$f(y) \leq f(x) + (y-x)^T \nabla f(x) + \frac{L}{2} \|y-x\|^2 \quad \forall x, y \in S$$

set $y = x_{k+1} \in S$

$x = x_k \in S$

$$f(x_{k+1}) \leq f(x_k) + (\alpha_k d_k)^T \nabla f(x_k) + \frac{L}{2} \alpha_k^2 \|d_k\|^2$$

$$\therefore x_{k+1} = x_k + \alpha_k d_k \quad (d_k = \bar{x}_k - x_k)$$

$$\Rightarrow f(x_{k+1}) \leq f(x_k) + \alpha_k (-g_k) + \frac{c}{2} \alpha_k^2$$

where $c = L \max_{\xi_1, \xi_2 \in S} \|\xi_1 - \xi_2\|^2$

finite

S $\rightarrow$ bounded closed

$$\begin{aligned} \|d_k\|^2 &= \|\bar{x}_k - x_k\|^2 \\ &\leq \max_{\xi_1, \xi_2 \in S} \|\xi_1 - \xi_2\|^2 \end{aligned}$$

Define a function $h(x) = f(x) - f(x^*)$

so $h(x_k) = f(x_k) - f(x^*) \leq g_k$
 (proved already)

$$\Rightarrow -h(x_k) \geq -g_k$$

$$\textcircled{1} \quad h(x_{k+1}) \leq f(x_k) - \alpha_k h(x_k) + \frac{c}{2} \alpha_k^2$$

as $\alpha_k > 0$

$$\Rightarrow h(x_{k+1}) \leq (1 - \alpha_k) h(x_k) + \frac{c}{2} \alpha_k^2$$

let $x_k = \frac{2}{k+2}$? why?

$$h(x_{k+1}) \leq \frac{k}{k+2} h(x_k) + \frac{2c}{(k+2)^2} \quad \text{--- (2)}$$

by induction on k , we can prove that

$$0 \leq h(x_k) \leq \frac{2\tilde{c}}{k+2}$$

for some constant $\tilde{c} > 0$

by sandwich theorem

$$\lim_{k \rightarrow \infty} h(x_k) = 0$$

$$\Rightarrow \lim_{k \rightarrow \infty} f(x_k) = f(x^*)$$

To prove induction part

$$k=0 \quad 0 \leq h(x_0) = f(x_0) - f(x^*)$$

$$\leq \max_{x \in S} f(x) - \min_{x \in S} f(x)$$



attained
(something)
finite

$$\Rightarrow 0 \leq h(x_0) \leq \text{some constant}$$

$$\text{--- (3)}$$

$$\text{let } h(x_k) \leq \frac{2\tilde{c}}{k+2}$$

$$\text{To prove } h(x_{k+1}) \leq \frac{2\tilde{c}}{k+3}$$

$$\text{consider } h(x_{k+1}) \leq \frac{k}{k+2} h(x_k) + \frac{2c}{(k+2)^2}$$

$$\leq \frac{k}{k+2} \cdot \frac{2\tilde{c}}{k+2} + \frac{2c}{(k+2)^2}$$

$$\left. \begin{array}{l} \text{set } \tilde{c} \geq \max \{c, \eta\} \\ \text{from (3)} \end{array} \right\} \leq \frac{2\tilde{c}k}{(k+2)^2} + \frac{2\tilde{c}}{(k+2)^2}$$

$$h(x_0) \leq \tilde{c} = \frac{2\tilde{c}}{0+2} \quad \text{[scribbled out]} \quad = \frac{2\tilde{c}}{(k+2)^2} (k+1)$$

$$\frac{k+1}{(k+2)^2} < \frac{1}{k+3}$$

Justification for $\alpha_k = \frac{k}{k+2}$??

$$\phi(\alpha) = (1-\alpha)h(x_k) + \frac{c}{2}\alpha^2$$

take: $\phi(\alpha) = (1-\alpha)h(x_k) + \frac{c}{2}\alpha^2$

$$\phi'(\alpha) = 0 \Rightarrow -h(x_k) + c\alpha = 0$$

$$\Rightarrow \alpha = \frac{h(x_k)}{c} \equiv \alpha_k$$

$$h(x_{k+1}) \leq \left(1 - \frac{h(x_k)}{c}\right) h(x_k) + \frac{c}{2} \frac{(h(x_k))^2}{c^2}$$

$$= h(x_k) \left(1 - \frac{h(x_k)}{2c}\right)$$

$$\frac{1}{h(x_{k+1})} \geq \frac{1}{h(x_k) \left(1 - \frac{h(x_k)}{2c}\right)}$$

$$\geq \frac{1}{h(x_k)} \left(1 + \frac{h(x_k)}{2c}\right)$$

$$\frac{1}{h(x_{k+1})} \geq \frac{1}{h(x_k)} + \frac{1}{2c}$$

$$\geq \frac{1}{h(x_{k+1})} + \frac{2}{2c} \geq \frac{1}{h(x_{k-2})} + \frac{3}{2c}$$

$$\frac{1}{h(x_{k+1})} \geq \frac{1}{h(x_0)} + \frac{k}{2c}$$

$$= \frac{2c + kh(x_0)}{2ch(x_0)}$$

$$\Rightarrow h(x_{k+1}) \leq \frac{2ch(x_0)}{2c + kh(x_0)} = \frac{2c}{k + \left(\frac{2c}{h(x_0)}\right)}$$

$$h(x_0) = f(x_0) - f(x^*) > 0$$

$\uparrow$ not known $\uparrow$ known $\uparrow$ not known